LESSON PLAN

EXPLOSIVE REGULATIONS (31)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	03
Total No. of Lesson Plans	4	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Applications of pyrotechnics store
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Applications of pyrotechnics store
5. Specific Objectives	31.1.1.1 Trainee will be able to explain the application of pyrotechnic stores

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of pyrotechnic store applications; Safety precautions regarding storage and handling; Instructor supervision of theoretical discussion
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary safety hazards associated with pyrotechnic stores?
(b) List three general categories of pyrotechnic devices used in aviation.
(c) Why is strict adherence to explosive regulations essential for aircrew?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Signal cartridges for visual communication • Smoke and flame generation for marking • Flotation devices for emergency signaling • Automatic markers for navigation aid • Marine depth charges for anti-submarine warfare	L & Ds	PPT, IB	15 min
(b)	• Operational procedures for signal cartridges • Deployment techniques for smoke and flame • Activation mechanisms of automatic markers • Safety protocols during depth charge practice • Maintenance and inspection of pyrotechnic stores	L & Ds	PPT, PDF	20 min
(c)	• Comparison of different pyrotechnic applications • Regulatory compliance for storage and transport • Emergency response to accidental ignition • Role of pyrotechnics in rescue operations • Future trends in pyrotechnic technology	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the specific application of a signal cartridge in an emergency scenario.
(b) Describe the safety precautions required when handling marine depth charges.
(c) How does an automatic marker function to assist in navigation?

12. Home Assignment and Follow Up	Prepare a brief report on the differences between signal cartridges and smoke flares, referencing the regulations covered in LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Construction and function of – Signal cartridge Smoke and flame float Automatic marker marine 110min OMAB 25 12D/8N Practice depth charge. Power cartridges	Refer to the master syllabus reference list	LP 04

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min